1) a)

$x = r\cos\theta$ and $y = r\sin\theta$

1) b)

$r = \sqrt{x^2 + y^2}$ and $\theta = \arctan\left(\frac{y}{x}\right)$

2) a)

$\sqrt{x^2 + y^2} = 3$ or $x^2 + y^2 = 9$

2) b)

$y = 5$

2) c)

$x = 2$

2) d)

$r = 4 + 4\cos\theta$

Multiply both sides by r:

$r^2 = 4r + 4r\cos\theta$

$x^2 + y^2 = 4\sqrt{x^2 + y^2} + 4x$

2) e)

$r = 2 - 2\sin\theta$

Multiply both sides by r:

$r^2 = 2r - 2r\sin\theta$

$x^2 + y^2 = 2\sqrt{x^2 + y^2} - 2y$

2) f)

$r + r\cos\theta = 4$

$\sqrt{x^2 + y^2} + x = 4$

2) g)

Multiply through by r:

$r^2 = 4r\cos\theta + 2r\sin\theta$

$x^2 + y^2 = 4x + 2y$

2) h)

Multiply through by r:

$r^2 = -2r\cos\theta$

$x^2 + y^2 = -2x$

2) i)

$3r + r\sin\theta = 1$

$3\sqrt{x^2 + y^2} + y = 1$

2) j)

Multiply through by r:

$r = \tan\theta$

$r = \frac{r\sin\theta}{r\cos\theta}$

$\sqrt{x^2 + y^2} = \frac{y}{x}$

3) a)

$r^2 = 25$

So $r = 5$

3) b)

$r^2 = 4r\cos\theta$

$r = 4\cos\theta$

3) c)

$r\cos\theta \times r\sin\theta = 9$

$r^2 = \frac{9}{\cos\theta\sin\theta}$

$r = \sqrt{\frac{9}{\cos\theta\sin\theta}}$

3) d)

$r\sin\theta = (r\cos\theta)^2 - 3r\cos\theta$

$\sin\theta = r\cos^2\theta - 3\cos\theta$

$\sin\theta + 3\cos\theta = r\cos^2\theta$

$r = \frac{\sin\theta + 3\cos\theta}{\cos^2\theta}$

3) e)

$\frac{(r\cos\theta)^2}{16} + \frac{(r\sin\theta)^2}{9} = 1$

$\frac{r^2\cos^2\theta}{16} + \frac{r^2\sin^2\theta}{9} = 1$

$9r^2\cos^2\theta + 16r^2\sin^2\theta = 144$

$r^2(9\cos^2\theta + 16\sin^2\theta) = 144$

$r^2 = \frac{144}{9\cos^2\theta + 16\sin^2\theta}$

$r = \sqrt{\frac{144}{9\cos^2\theta + 16\sin^2\theta}}$

3) f)

$(r^2)^2 = (r\cos\theta)^2 - (r\sin\theta)^2$

$r^4 = r^2\cos^2\theta - r^2\sin^2\theta$

$r^2 = \cos^2\theta - \sin^2\theta$

$r = \sqrt{\cos^2\theta - \sin^2\theta}$

3) g)

$(r\cos\theta)^2(r\sin\theta) = 6$

$r^3\cos^2\theta\sin\theta = 6$

$r = \sqrt[3]{\frac{6}{\cos^2\theta\sin\theta}}$

4) a)

Multiply through by r:

$r^2 = 2r\cos\theta - 5r\sin\theta$

$x^2 + y^2 = 2x - 5y$

$x^2 + y^2 - 2x + 5y = 0$

$(x-1)^2 + \left(y + \frac{5}{2}\right)^2 - 1 - \frac{25}{4} = 0$

$(x-1)^2 + \left(y + \frac{5}{2}\right)^2 = \frac{29}{4}$

which is an equation of a circle.

4) b)

$\left(1, -\frac{5}{2}\right)$

4) c)

$r = \sqrt{\frac{29}{4}} = \frac{\sqrt{29}}{2}$

5)

$(r\cos\theta)^2 + (r\sin\theta)^2 + 4r\sin\theta = 12$

$r^2\cos^2\theta + r^2\sin^2\theta + 4r\sin\theta = 12$

$r^2(\cos^2\theta + \sin^2\theta) + 4r\sin\theta = 12$

$r^2 + 4r\sin\theta = 12$

$(r + 2\sin\theta)^2 - 4\sin^2\theta = 12$

$(r + 2\sin\theta)^2 = 12 + 4\sin^2\theta$

$r = -2\sin\theta \pm \sqrt{12 + 4\sin^2\theta}$

Since $r > 0$

$r = -2\sin\theta + \sqrt{12 + 4\sin^2\theta}$ as req.

6) a)

Multiply through by r:

$r^2 = ar\cos\theta$

$x^2 + y^2 = ax$

6) b)

Multiply through by r:

$r^2 = ar\sin\theta$

$x^2 + y^2 = ay$

6) c)

Multiply through by r:

$r^2 = a(r + r\cos\theta)$

$x^2 + y^2 = a(\sqrt{x^2 + y^2} + x)$

6) d)

Multiply through by r:

$r^2 = a(r - r\sin\theta)$

$x^2 + y^2 = a(\sqrt{x^2 + y^2} - y)$

7) a)

Multiply through by r:

$r^2 = a(r\cos\theta + r\sin\theta)$

$x^2 + y^2 = a(x + y)$

$x^2 + y^2 - ax - ay = 0$

$\left(x - \frac{a}{2}\right)^2 + \left(y - \frac{a}{2}\right)^2 - 2 \times \left(\frac{a}{2}\right)^2 = 0$

$\left(x - \frac{a}{2}\right)^2 + \left(y - \frac{a}{2}\right)^2 = \frac{a^2}{2}$

Radius is $\frac{a}{\sqrt{2}} = \frac{\sqrt{2}}{2}a$

7) b)

Centre at $\left(\frac{a}{2}, \frac{a}{2}\right)$

8) a)

Multiply through by r:

$r^2 = a(pr\cos\theta + qr\sin\theta)$

$x^2 + y^2 = a(px + qy)$

$x^2 + y^2 - apx - aqy = 0$

$\left(x - \frac{ap}{2}\right)^2 + \left(y - \frac{aq}{2}\right)^2 - \left(\frac{ap}{2}\right)^2 - \left(\frac{aq}{2}\right)^2 = 0$

$\left(x - \frac{ap}{2}\right)^2 + \left(y - \frac{aq}{2}\right)^2 = \frac{a^2}{4}(p^2 + q^2)$

8) b)

Centre at $\left(\frac{ap}{2}, \frac{aq}{2}\right)$

8) c)

Radius is $\frac{a}{2}\sqrt{p^2 + q^2}$

9) a)

Centre at $(8, -6)$, radius 10

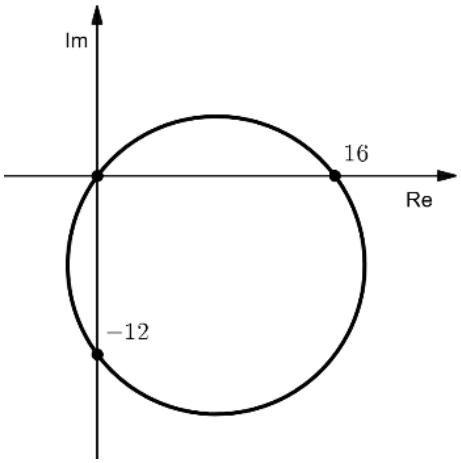
$(x - 8)^2 + (y + 6)^2 = 100$

When $x = 0$:

$(y + 6)^2 = 36 \Rightarrow y = -6 \pm 6$ so $(0, 0)$ and $(0, -12)$

When $y = 0$:

$(x - 8)^2 = 64 \Rightarrow x = 8 \pm 8$ so $(16, 0)$ and $(0, 0)$



9) b)

$(x-8)^2+(y+6)^2=100$

$(r\cos\theta-8)^2+(r\sin\theta+6)^2=100$

$r^2\cos^2\theta-16r\cos\theta+64+r^2\sin^2\theta+12r\sin\theta+36=100$

$r^2(\cos^2\theta+\sin^2\theta)-16r\cos\theta+12r\sin\theta=0$

$r^2=16r\cos\theta-12r\sin\theta$

Since $r\neq 0$

$r=16\cos\theta-12\sin\theta$

10) a)

Centre at $(-2,4)$, radius $\sqrt{20}$

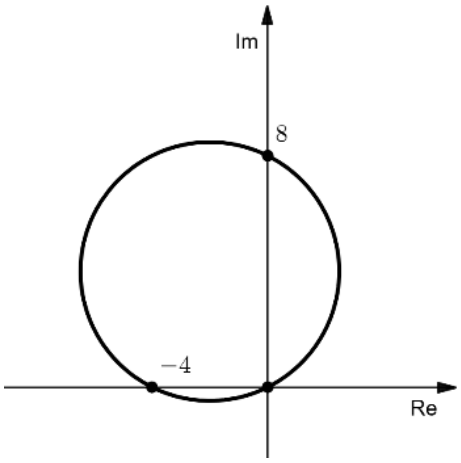
$(x+2)^2+(y-4)^2=20$

When $x=0$:

$(y-4)^2=16\Rightarrow y=4\pm 4$ so $(0,8)$ and $(0,0)$

When $y=0$:

$(x+2)^2=4\Rightarrow x=-2\pm 2$ so $(0,0)$ and $(-4,0)$



10) b)

$(x+2)^2+(y-4)^2=20$

$(r\cos\theta+2)^2+(r\sin\theta-4)^2=20$

$r^2\cos^2\theta+4r\cos\theta+4+r^2\sin^2\theta-8r\sin\theta+16=20$

$r^2(\cos^2\theta+\sin^2\theta)+4r\cos\theta-8r\sin\theta=0$

$r^2=8r\sin\theta-4r\cos\theta$

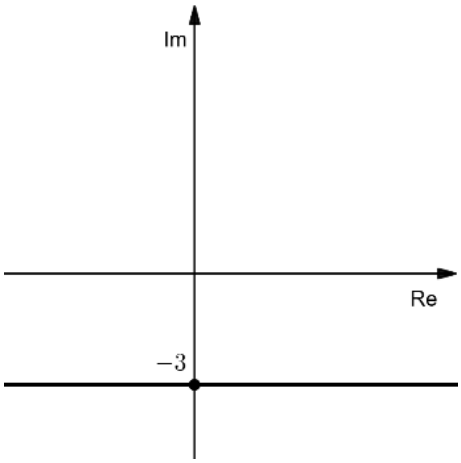
Since $r\neq 0$

$r=8\sin\theta-4\cos\theta$

11) a)

Perpendicular bisector of $(0, 4)$ and $(0, -10)$

Midpoint: $(0, -3)$ so $y = -3$



11) b)

$y = -3$

$r \sin\theta = -3$

$r = -\frac{3}{\sin\theta}$

12) a)

$\sqrt{x^2 + y^2} = \frac{1}{\sqrt{1}}$

$x^2 + y^2 = 1$

12) b)

$r^2 = \frac{16}{16\cos^2\theta + \sin^2\theta}$

$1 = \frac{16}{16r^2\cos^2\theta + r^2\sin^2\theta}$

$1 = \frac{16}{16x^2 + y^2}$

$16x^2 + y^2 = 16$

$x^2 + \frac{y^2}{16} = 1$

12) c)

$$r^2 = \frac{400}{25\cos^2\theta + 16\sin^2\theta}$$
$$1 = \frac{400}{25r^2\cos^2\theta + 16r^2\sin^2\theta}$$
$$1 = \frac{400}{25x^2 + 16y^2}$$
$$25x^2 + 16y^2 = 400$$
$$\frac{x^2}{16} + \frac{y^2}{25} = 1$$

12) d)

$$r^2 = \frac{36}{9\cos^2\theta + 4\sin^2\theta}$$
$$1 = \frac{36}{9r^2\cos^2\theta + 4r^2\sin^2\theta}$$
$$1 = \frac{36}{9x^2 + 4y^2}$$
$$9x^2 + 4y^2 = 36$$
$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$

13) a)

$$\frac{(r\cos\theta)^2}{4} + \frac{(r\sin\theta)^2}{25} = 1$$
$$25r^2\cos^2\theta + 4r^2\sin^2\theta = 100$$
$$r^2(25\cos^2\theta + 4\sin^2\theta) = 100$$
$$r^2 = \frac{100}{25\cos^2\theta + 4\sin^2\theta}$$
$$r = \frac{10}{\sqrt{25\cos^2\theta + 4\sin^2\theta}}$$

13) b)

$$\frac{(r\cos\theta)^2}{16} + \frac{(r\sin\theta)^2}{9} = 1$$
$$9r^2\cos^2\theta + 16r^2\sin^2\theta = 144$$
$$r^2(9\cos^2\theta + 16\sin^2\theta) = 144$$
$$r^2 = \frac{144}{9\cos^2\theta + 16\sin^2\theta}$$
$$r = \frac{12}{\sqrt{9\cos^2\theta + 16\sin^2\theta}}$$

13) c)

$\frac{9}{16}(r\cos\theta)^2 + \frac{49}{64}(r\sin\theta)^2 = 1$

$36r^2\cos^2\theta + 49r^2\sin^2\theta = 64$

$r^2(36\cos^2\theta + 49\sin^2\theta) = 64$

$r^2 = \frac{64}{36\cos^2\theta + 49\sin^2\theta}$

$r = \frac{8}{\sqrt{36\cos^2\theta + 49\sin^2\theta}}$

14) a)

$-1 \leq \sin\theta \leq 1$

So max of $r = \frac{6}{4-2} = \frac{6}{2} = 3$

14) b)

$-1 \leq \sin\theta \leq 1$

So min of $r = \frac{6}{4+2} = \frac{6}{6} = 1$

14) c)

$r(4 + 2\sin\theta) = 6$

$4r + 2r\sin\theta = 6$

$4\sqrt{x^2 + y^2} + 2y = 6$

$\sqrt{x^2 + y^2} = \frac{6-2y}{4}$

$x^2 + y^2 = \left(\frac{6-2y}{4}\right)^2$

$x^2 = \left(\frac{6-2y}{4}\right)^2 - y^2$

$x^2 = \frac{36-24y+4y^2}{16} - y^2$

$x^2 = \frac{9}{4} - \frac{3}{2}y - \frac{3}{4}y^2$

14) d)

$x^2 = \frac{9}{4} - \frac{3}{2}y - \frac{3}{4}y^2$

$\frac{4}{3}x^2 = 3 - 2y - y^2$

$\frac{4}{3}x^2 = 3 - (y+1)^2 + 1$

$\frac{4}{3}x^2 = -(y+1)^2 + 4$

$\frac{x^2}{3} + \frac{(y+1)^2}{4} = 1$

15) a)

$-1 \leq \sin\theta \leq 1$

So max of $r = \frac{16}{3-1} = \frac{16}{2} = 8$

15) b)

$-1 \leq \sin\theta \leq 1$

So min of $r = \frac{16}{3+1} = \frac{16}{4} = 4$

15) c)

$r(3 - \sin\theta) = 16$

$3r - r\sin\theta = 16$

$3\sqrt{x^2+y^2} - y = 16$

$\sqrt{x^2+y^2} = \frac{16+y}{3}$

$x^2 + y^2 = \left(\frac{16+y}{3}\right)^2$

$x^2 = \left(\frac{16+y}{3}\right)^2 - y^2$

$x^2 = \frac{256+32y+y^2}{9} - y^2$

$x^2 = \frac{256}{9} + \frac{32}{9}y - \frac{8}{9}y^2$

15) d)

$\frac{9}{8}x^2 = 32 + 4y - y^2$

$\frac{9}{8}x^2 = 32 - (y-2)^2 + 4$

$\frac{9}{8}x^2 = -(y-2)^2 + 36$

$\frac{x^2}{32} + \frac{(y-2)^2}{36} = 1$

16)

$r = \frac{k}{2+\cos\theta}$

$2r + r\cos\theta = k$

$2\sqrt{x^2+y^2} + x = k$

$\sqrt{x^2+y^2} = \frac{k-x}{2}$

$x^2 + y^2 = \left(\frac{k-x}{2}\right)^2$

$y^2 = \frac{1}{4}k^2 - \frac{1}{2}kx + \frac{1}{4}x^2 - x^2$

$y^2 = \frac{1}{4}k^2 - \frac{1}{2}kx - \frac{3}{4}x^2$

$\frac{4}{3}y^2 = \frac{1}{3}k^2 - \frac{2}{3}kx - x^2$

$\frac{4}{3}y^2 = \frac{1}{3}k^2 - \left(x + \frac{1}{3}k\right)^2 + \frac{1}{9}k^2$

$\frac{4}{3}y^2 + \left(x + \frac{1}{3}k\right)^2 = \frac{4}{9}k^2$

$\frac{\left(x+\frac{1}{3}k\right)^2}{\frac{4}{9}k^2} + \frac{y^2}{\frac{1}{3}k^2} = 1$

17) a)

$r^2\sin 2\theta = 9 \Rightarrow 2r^2\sin\theta\cos\theta = 9$

$2xy = 9$

17) b)

$r^2\cos 2\theta = 4 \Rightarrow r^2(\cos^2\theta - \sin^2\theta) = 4$

$r^2\cos^2\theta - r^2\sin^2\theta = 4$

$x^2 - y^2 = 4$

17) c)

$r^4 \sin 4\theta = 4 \Rightarrow 2r^4 \sin 2\theta \cos 2\theta = 4$

$4r^4 \sin\theta \cos\theta \,(\cos^2\theta - \sin^2\theta) = 4$

$r^2 \sin\theta \cos\theta \,(r^2\cos^2\theta - r^2\sin^2\theta) = 1$

$xy(x^2 - y^2) = 1$

18)

$r^4 \cos 4\theta = 1 \Rightarrow r^4(\cos^2 2\theta - \sin^2 2\theta) = 1$

$r^4((\cos 2\theta)^2 - (\sin 2\theta)^2) = 1$

$r^4((\cos^2\theta - \sin^2\theta)^2 - (2\sin\theta\cos\theta)^2) = 1$

$r^4(\cos^4\theta - 2\cos^2\theta\sin^2\theta + \sin^4\theta - 4\sin^2\theta\cos^2\theta) = 1$

$r^4\cos^4\theta - 2r^4\cos^2\theta\sin^2\theta + r^4\sin^4\theta - 4r^4\sin^2\theta\cos^2\theta = 1$

$x^4 - 2x^2y^2 + y^4 - 4x^2y^2 = 1$

$x^4 - 6x^2y^2 + y^4 = 1$

19) a)

$(r\cos\theta)^4 + (r\sin\theta)^4 = (r\cos\theta)^2 + (r\sin\theta)^2$

$r^4\cos^4\theta + r^4\sin^4\theta = r^2\cos^2\theta + r^2\sin^2\theta$

$r^2\cos^4\theta + r^2\sin^4\theta = \cos^2\theta + \sin^2\theta$

$r^2\cos^4\theta + r^2\sin^4\theta = 1$

$r^2(\cos^4\theta + \sin^4\theta) = 1$

Since

$\cos^4\theta + \sin^4\theta = (\cos^2\theta + \sin^2\theta)^2 - 2\cos^2\theta\sin^2\theta$

$= 1 - 2\cos^2\theta\sin^2\theta$

$= 1 - 2(\sin\theta\cos\theta)(\sin\theta\cos\theta)$

$= 1 - \frac{1}{2}(2\sin\theta\cos\theta)(2\sin\theta\cos\theta)$

$= 1 - \frac{1}{2}\sin^2 2\theta$

Therefore

$r^2(\cos^4\theta + \sin^4\theta) = 1$ gives

$r^2 = \frac{1}{\cos^4\theta + \sin^4\theta} = \frac{1}{1 - \frac{1}{2}\sin^2 2\theta}$

So that

$r^2 = \frac{2}{2 - \sin^2 2\theta}$

19) b)

Since $0 \leq \sin^2 2\theta \leq 1$

The maximum r can be is when $\sin^2 2\theta = 1$

$\sin 2\theta = 1$ has solutions $\theta = \frac{\pi}{4}$ and $\theta = \frac{5\pi}{4}$

$\sin 2\theta = 1$ has solutions $\theta = \frac{3\pi}{4}$ and $\theta = \frac{7\pi}{4}$

$r^2 = \frac{2}{2-1} = 2$

The maximum of r is $\sqrt{2}$

So the points are $\left(\sqrt{2}, \frac{\pi}{4}\right)$, $\left(\sqrt{2}, \frac{3\pi}{4}\right)$, $\left(\sqrt{2}, \frac{5\pi}{4}\right)$ and $\left(\sqrt{2}, \frac{7\pi}{4}\right)$

20) a)

$(r\cos\theta)^3 + (r\sin\theta)^3 = 12(r\cos\theta)(r\sin\theta)$

$r^3\cos^3\theta + r^3\sin^3\theta = 12r^2\cos\theta\sin\theta$

$r\cos^3\theta + r\sin^3\theta = 12\cos\theta\sin\theta$

$r = \frac{12\cos\theta\sin\theta}{\cos^3\theta + \sin^3\theta}$

20) b)

$12\cos\theta\sin\theta = 0$

So either $\cos\theta = 0$ or $\sin\theta = 0$

So that $\theta \in \left\{\frac{\pi}{2}, \pi\right\}$

20) c)

$r = \frac{12\cos\left(\frac{\pi}{4}\right)\sin\left(\frac{\pi}{4}\right)}{\cos^3\left(\frac{\pi}{4}\right)+\sin^3\left(\frac{\pi}{4}\right)} = \frac{12\times\frac{\sqrt{2}}{2}\times\frac{\sqrt{2}}{2}}{\left(\frac{\sqrt{2}}{2}\right)^3+\left(\frac{\sqrt{2}}{2}\right)^3} = \frac{6}{\frac{2\sqrt{2}}{8}+\frac{2\sqrt{2}}{8}} = \frac{12}{\sqrt{2}} = 6\sqrt{2}$

20) d)

The line goes through polar coordinate $\left(6\sqrt{2}, \frac{\pi}{4}\right)$ and the pole.

$\therefore$ line of symmetry is $y = \tan\left(\frac{\pi}{4}\right)x \Rightarrow y = x$

21) a)

$\theta = \pi$ so $r = 2 - 6\sin\pi = 2$

$(-2, 0)$

21) b)

When $\sin\theta = -1$ then $r = 2 + 6 = 8$ when $\theta = -\frac{\pi}{2}$

$\left(8, -\frac{\pi}{2}\right)$

21) c)

Curve C_2 is $r = 5$

So

$5 = 2 - 6\sin\theta$

$\sin\theta = \frac{1}{2}$ which gives $\theta = -\frac{5\pi}{6}$ or $\theta = -\frac{\pi}{6}$

In Cartesian coordinates:

$M\left(5\cos\left(-\frac{5\pi}{6}\right), 5\sin\left(-\frac{5\pi}{6}\right)\right)$ so $\left(-\frac{5\sqrt{3}}{2}, -\frac{5}{2}\right)$

$N\left(5\cos\left(-\frac{\pi}{6}\right), 5\sin\left(-\frac{5\pi}{6}\right)\right)$ so $\left(\frac{5\sqrt{3}}{2}, -\frac{5}{2}\right)$

$Q\left(8\cos\left(-\frac{\pi}{2}\right), 8\sin\left(-\frac{\pi}{2}\right)\right)$ so $(0, -8)$

So the area of the triangle MNQ is $\frac{1}{2} \times 2 \times \frac{5\sqrt{3}}{2} \times (8 - 2.5) = \frac{55\sqrt{3}}{4}$ units^2 as req.

22) a)

$3r + 2r\cos\theta = 1$

$3\sqrt{x^2 + y^2} + 2x = 1$

$3\sqrt{x^2 + y^2} = 1 - 2x$

$9(x^2 + y^2) = (1 - 2x)^2$

$9x^2 + 9y^2 = 1 - 4x + 4x^2$

$9y^2 = 1 - 4x - 5x^2$

$9y^2 = (1 - 5x)(1 + x)$

22) b)

Since the terms `$9y^2$`, `$-5x$` and `$+x$` remain in both curves, it is a translation in the x-direction.

Let $x' = x - 1$

$9y^2 = \big(1 - 5(x-1)\big)(1 + (x-1))$

$= (6 - 5x)x$

which is curve C_2 so the transformation is a translation by $\begin{pmatrix}1\\0\end{pmatrix}$

23) a)

$4r - 3r\cos\theta = 7$

$4\sqrt{x^2+y^2} - 3x = 7$

$4\sqrt{x^2+y^2} = 7 + 3x$

$16(x^2+y^2) = (7+3x)^2$

$16x^2 + 16y^2 = 9x^2 + 42x + 49$

$7x^2 - 42x + 16y^2 = 49$

$7(x^2 - 6x) + 16y^2 = 49$

$7((x-3)^2 - 9) + 16y^2 = 49$

$7(x-3)^2 + 16y^2 = 112$

$\frac{(x-3)^2}{16} + \frac{y^2}{7} = 1$

$\therefore a = 16$ and $b = 7$

23) b)

Translation $\begin{pmatrix}-3\\0\end{pmatrix}$

24)

Multiply through by r:

$r^2 = r\cos\theta - \frac{r}{\sin\theta}$

$r^2 = r\cos\theta - \frac{r^2}{r\sin\theta}$

$x^2 + y^2 = x - \frac{x^2+y^2}{y}$

$x^2y + y^3 = xy - x^2 - y^2$

So the transformation is a reflection in the line $y = x$

25) a)

Curve C_1 in polar form:

$x^2 + y^2 - x = \sqrt{x^2 + y^2}$

$r^2 - r\cos\theta = r$

$r - \cos\theta = 1$

$r = 1 + \cos\theta$

Solve simultaneously:

$1 - \sin\theta = 1 + \cos\theta$

$-\sin\theta = \cos\theta$

$\frac{\sin\theta}{\cos\theta} = -1$

$\tan\theta = -1$

$\theta = \left\{\frac{3\pi}{4}, \frac{7\pi}{4}\right\}$

So $r = 1 - \sin\frac{3\pi}{4} = 1 - \frac{\sqrt{2}}{2}$ and $r = 1 - \sin\frac{7\pi}{4} = 1 + \frac{\sqrt{2}}{2}$

$\left(1 - \frac{\sqrt{2}}{2}\right)\cos\frac{3\pi}{4} = \left(1 - \frac{\sqrt{2}}{2}\right)\left(-\frac{\sqrt{2}}{2}\right) = \frac{1}{2} - \frac{\sqrt{2}}{2}$

$\left(1 - \frac{\sqrt{2}}{2}\right)\sin\frac{3\pi}{4} = \left(1 - \frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{2} - \frac{1}{2}$

$\left(1 + \frac{\sqrt{2}}{2}\right)\cos\frac{7\pi}{4} = \left(1 + \frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{2} + \frac{1}{2}$

$\left(1 + \frac{\sqrt{2}}{2}\right)\sin\frac{7\pi}{4} = \left(1 + \frac{\sqrt{2}}{2}\right)\left(-\frac{\sqrt{2}}{2}\right) = -\frac{\sqrt{2}}{2} - \frac{1}{2}$

So $\left(\frac{1}{2} - \frac{\sqrt{2}}{2}, -\frac{1}{2} + \frac{\sqrt{2}}{2}\right)$ and $\left(\frac{1}{2} + \frac{\sqrt{2}}{2}, -\frac{1}{2} - \frac{\sqrt{2}}{2}\right)$

25) b)

$y = -x$

26)

To find N: When $\cos\theta = 1$ so $r = \frac{52}{129-100} = \frac{52}{29}$

So as cartesian coordinates: $N\left(\frac{52}{29}, 0\right)$

To find P and Q solve simultaneously:

$r = \frac{52}{129-100r^2}$

$r(129 - 100r^2) = 52$

$100r^3 - 129r - 52 = 0$

$(2r - 1)(5r - 4)(10r + 13) = 0$

$r = \frac{1}{2}$, $r = \frac{4}{5}$ or $r = -\frac{13}{10}$

Since $r > 0$, $r = \frac{1}{2}$, $r = \frac{4}{5}$

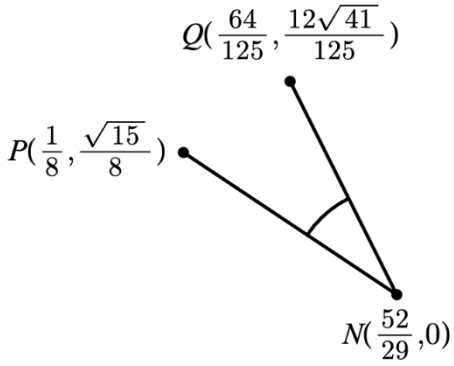
When $r = \frac{1}{2}$

$\frac{1}{4} = \cos\theta$ so $\theta = 1.31811607$ ...

When $r = \frac{4}{5}$

$\frac{16}{25} = \cos\theta$ so $\theta = 0.8762980612$ ...

So as Cartesian coordinates: $P\left(\frac{1}{8}, \frac{\sqrt{15}}{8}\right)$ and $Q\left(\frac{64}{125}, \frac{12\sqrt{41}}{125}\right)$



Side NP : $\sqrt{\left(\frac{52}{29} - \frac{1}{8}\right)^2 + \left(0 - \frac{\sqrt{15}}{8}\right)^2} = \frac{\sqrt{10149}}{58}$

Side NQ : $\sqrt{\left(\frac{52}{29} - \frac{64}{125}\right)^2 + \left(0 - \frac{12\sqrt{41}}{125}\right)^2} = \frac{12\sqrt{7370}}{725}$

Side PQ : $\sqrt{\left(\frac{64}{125} - \frac{1}{8}\right)^2 + \left(\frac{12\sqrt{41}}{125} - \frac{\sqrt{15}}{8}\right)^2} = \sqrt{\frac{381-12\sqrt{615}}{500}}$

By the cosine rule:

$\cos\theta = \frac{NP^2+NQ^2-PQ^2}{2\times NP\times NQ} = 0.986433$ ...

So $\theta = 9.45°$ (3 SF)